

VibMotion: Designing Programmable Vibration-Driven Locomotion via 3D Printed Hook Structures and Stick-Slip Effects

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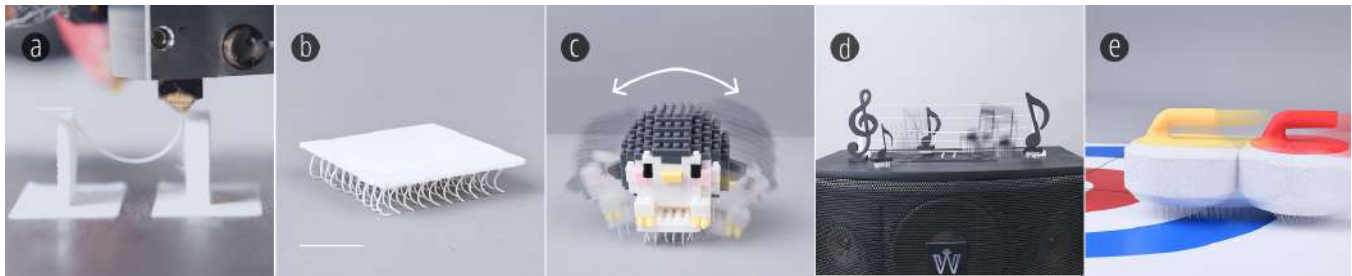


Figure 1: VibMotion System and Applications (a) Hook-shaped Fabrication Pipeline, (b) VibMotion standard module, (c) Lego-Compatible Components, (d) Everyday Vibration as Expressive Motion, (e) Embodied Motion Play. Scale Bar: 20mm.

Abstract

We present VibMotion, which enables programmable locomotion under vibration excitation through customizable 3D-printed structures. VibMotion employs a G-code-controlled 3D printing process to fabricate directional, asymmetric hook-shaped geometries that leverage surface interactions to translate vibration excitation into stable motion behaviors without complex control. Through exploratory studies on a standardized motion module, we investigate how geometric orientation and arrangement influence locomotion outcomes under vibration, demonstrating multiple motion patterns including directional translation, turning, rotation, and hopping. By modularly combining these motion units, more complex motion behaviors can be composed from simple structural elements, expanding the design space of programmable morphology for interactive motion systems and playful embodied interaction.

CCS Concepts

• **Human-centered computing** → **Interactive system and tools.**

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Keywords

Vibration-driven locomotion, Programmable morphology, 3D printing

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1 Introduction

Vibration is a physical phenomenon that is ubiquitous in everyday environments, ranging from household appliances to mobile devices [6, 7]. Beyond being perceived as noise or disturbance, vibration has long been used as a simple means to drive motion, often requiring minimal actuation structures [2, 8]. Classic examples such as brush robots demonstrate that substantial locomotion can be achieved using only a single vibration motor, without wheels, joints, or complex transmission mechanisms [1]. Vibration-driven micro-robotic platforms and related design considerations have also been studied in robotics, highlighting both controllability opportunities and challenges [10, 11]. Compared to traditional actuation methods based on gears, cables, or pneumatic systems, vibration-driven approaches offer advantages in structural simplicity and fabrication accessibility [6, 8].

However, within human-computer interaction (HCI), vibration-driven locomotion has rarely been treated as a programmable design



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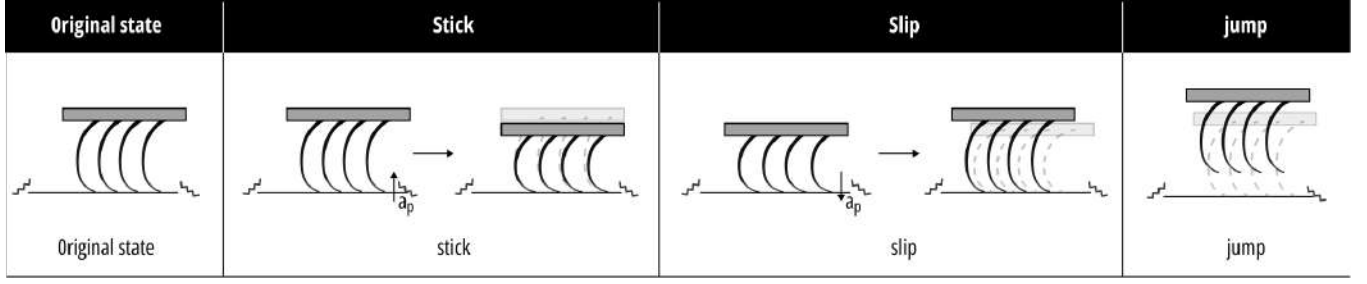


Figure 2: Motion principle

medium. Prior work such as CARDinality demonstrates expressive vibration-driven motion but frames locomotion as inherently stochastic, relying on training procedures or feedback systems to compensate for uncertainty [14]. In contrast, Cillia achieves directional motion through frictionally anisotropic microstructures and stick-slip effects, yet depends on precise micro-scale fabrication and complex manufacturing processes, raising barriers for broader adoption [5]. More broadly, HCI has argued for viewing material and morphology as programmable media [3], and adjacent fields such as soft robotics show how morphology can encode diverse behaviors [9]. Recent fabrication research further expands what can be encoded through 3D printing of hair-like structures and constructive platforms [12, 13]. As a result, existing approaches either rely on complex fabrication to achieve directionality or depend on sensing and control to manage motion uncertainty.

This gap raises a key question for HCI: how can programmable directional motion be achieved while preserving the structural simplicity and accessibility of vibration-driven actuation?

In this paper, we present *VibMotion*, a fabrication-driven approach to vibration-driven locomotion that enables programmable directional motion through customizable 3D-printed structures. VibMotion leverages G-code control over printing speed and extrusion within a standard FDM printing process to fabricate directional, arrayed hook-shaped structures on the surface of a base plate. Under vibration excitation, these structures interact with the supporting surface through stick-slip effects and asymmetric friction, resulting in stable and programmable directional locomotion.

2 VibMotion Overview

2.1 Fabrication Pipeline

VibMotion structures are fabricated using a G-code-controlled FDM printing process [4, 12] with PolyMax PLA, which provides sufficient elasticity for deformation under vibration while maintaining structural rigidity. The process begins by printing a flat base plate. After reaching the target height, the nozzle moves laterally while continuously extruding material (Fig. 3a). By increasing the extrusion rate and reducing the nozzle travel speed, the extruded filament temporarily suspends and sags under gravity before solidification (Fig. 3b), forming smooth hook-shaped geometries (Fig. 3c). This suspended extrusion is repeated layer by layer, followed by simple post-processing trimming to produce hook structures with consistent bending orientation (Fig. 3d).

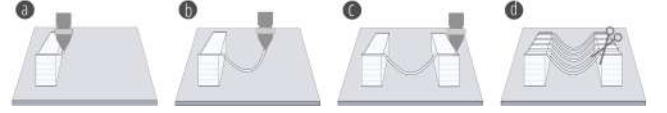


Figure 3: Fabrication pipeline

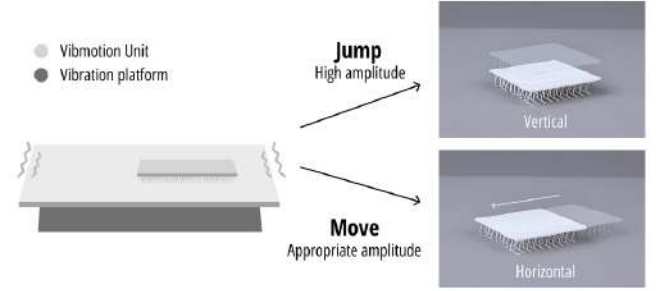


Figure 4: Vibration experiment

2.2 Motion Principle

To explore a wider range of locomotion behaviors, we employ a vibration platform that provides stable excitation. Prior to experimentation, we define a standardized VibMotion prototype (44 mm × 44 mm footprint, 100 hook elements, 1.2 mm base thickness, 10 mm hook length, total mass 3.4 g). Experiments reveal that within a moderate excitation range, the structure exhibits consistent directional motion driven by stick-slip effects (Fig. 4). When the vibration intensity becomes excessive, increased excitation amplifies inertial effects, leading to intermittent loss of contact and a transition toward jumping behavior.

Stick-Slip. Under vibration excitation, periodic vertical acceleration modulates the normal force at the contact interface between the hook structures and the vibrating surface, resulting in alternating sticking and slipping phases within each cycle. During upward acceleration, increased normal force strengthens frictional constraints, causing the asymmetric hook structures to elastically deform without net sliding. During downward acceleration, reduced normal force weakens frictional constraints, allowing the stored elastic energy to be released. Due to geometric asymmetry, slipping preferentially occurs along a permitted direction, producing a

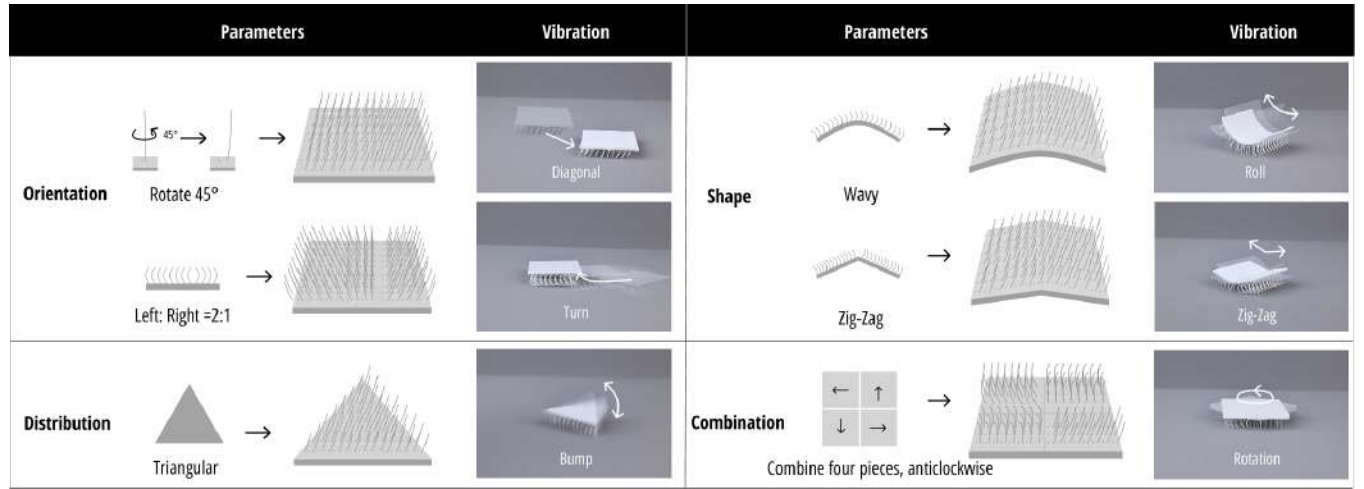


Figure 5: Motion pattern

net displacement per cycle. (Fig. 2) Repetition of this asymmetric stick-slip process yields stable directional locomotion.

Jump. As vibration amplitude further increases, downward acceleration becomes sufficient to temporarily reduce the normal force to near zero, causing intermittent loss of contact. In this regime, inertial effects dominate the dynamics, and the structure undergoes repeated detachment and collision with the surface, resulting in vertical jumping motion. (Fig. 2) Asymmetric deformation and contact conditions introduce a lateral velocity component at takeoff, giving rise to weakly directional jumping behavior.

2.3 Motion Pattern

By adjusting the parameters of the hook structures and the geometry of the base plate—including the bending orientation of the hooks, their spatial distribution, and the shape of the supporting surface—VibMotion enables a wide range of locomotion behaviors based on stick-slip effects. Variations in hook orientation and asymmetry can induce directional bias, resulting in motion patterns such as diagonal translation and turning. Modifying the distribution of hooks or introducing non-uniform base geometries further leads to behaviors such as bumping and rolling. In addition, shaping the base plate with wavy or zig-zag profiles allows the vibration-driven system to follow curved or segmented trajectories under excitation.

Beyond individual modules, VibMotion also supports user-driven composition by combining multiple modules with different geometric configurations. The superposition of motion tendencies across modules enables more complex locomotion outcomes, such as rotational motion when multiple units are arranged in an anticlockwise configuration. Together, these results demonstrate how controllable stick-slip interactions, encoded through fabrication parameters and modular composition, expand the design space of vibration-driven locomotion.

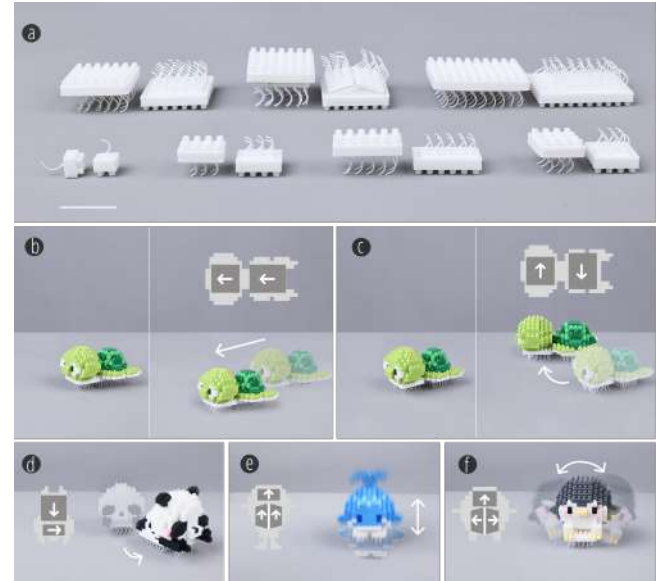


Figure 6: Lego-Compatible Components: (a) Parts compatible with different LEGO sizes. (b) and (c) the same object is equipped with components representing different directions, and when combined, they present different motion patterns. (d-f) other interesting motion patterns. Scale Bar: 20mm

3 Applications

3.1 Lego-Compatible Components

VibMotion base plate is designed as a modular component compatible with standard LEGO interfaces (Fig. 6), allowing it to be directly integrated into existing LEGO or building block systems. Under vibration excitation, different geometric configurations and module combinations exhibit distinct locomotion behaviors, enhancing the playfulness and expressive motion of LEGO-based constructions.



Figure 7: Embodied Motion Play:(a) VibCurling components, (b) assembly, (c) changing motion speed by reducing friction, (d) motion and collision.



Figure 8: Vibmotion in Life: (a) 3D printers, (b) Speakers.

3.2 Embodied Motion Play

We developed VibCurling (Fig. 7), in which the same vibration actuation can produce different motion trajectories and strategic behaviors by adjusting geometric parameters. Users can also modify surface friction through actions such as “sweeping,” thereby influencing the curling stone’s speed, trajectory stability, and collision outcomes. This translates curling into a vibration-driven motion experience that is perceptible and participatory.

3.3 Vibmotion in Life

VibMotion repurposes daily sources of vibration to create engaging motion in familiar environments. For example, the inherent vibration of a 3D printer during fabrication can cause printed objects to move immediately after printing, even producing playful effects such as “wagging” (Fig. 8a) or subtle oscillatory motion. Similarly, when placed on a speaker, VibMotion modules respond to music-driven vibration, animating objects such as musical notes and enhancing the visual expressiveness of sound (Fig. 8b). By embedding motion into existing vibration contexts, VibMotion introduces moments of playfulness and interaction into daily and workplace settings without requiring dedicated actuation systems.

4 Limitations and Discussions

Motion precision and repeatability. Under laboratory conditions, VibMotion achieves over 95% motion accuracy for a given geometry,

but performance degrades under varying surface friction, vibration stability, and repeated reuse. After approximately 50 trials, hook structures may detach or become misaligned. Future work will improve durability and repeatability through optimized printing logic and parameter control.

Load capacity and scalability. Current prototypes are fabricated using PLA-based materials, which limits load-bearing capacity due to deformation under higher loads. As VibMotion relies on vibration-driven stick-slip mechanisms, increased mass leads to greater energy dissipation and reduced motion efficiency, constraining achievable scale and application scenarios.

Limited post-fabrication reconfigurability. Although VibMotion supports diverse motion behaviors through geometric design, motion patterns cannot be altered within a single module after fabrication. Our current approach relies on modular composition to combine different motion tendencies. Future work could incorporate simple electronic mechanisms to enable dynamic switching and richer motion behaviors.

5 Conclusion

In this paper, we presented VibMotion, which enables programmable vibration-driven locomotion by encoding directional bias into asymmetric, hook-shaped 3D-printed contact geometries and leveraging stick-slip interactions with the supporting surface. Using a standardized module and controlled vibration excitation, we demonstrated stable directional motion in moderate operating conditions and a transition to jumping under higher excitation, and showed how adjusting geometric parameters yields multiple motion patterns such as translation, turning, rotation, and hopping. We further demonstrated through interesting applications that modular assembly enables users to combine the motion trends of different units to create more diverse composite behaviors. Together, VibMotion highlights a practical path for treating vibration as a designable motion resource and for expanding the demonstration and exploration space of motion-enabled interactive artifacts.

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